



R2.3 How far? The extent of chemical change (HL)

Reactivity 2 · How much, how fast and how far? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Many reactions are reversible and reach a state of dynamic equilibrium, where the forward and reverse reaction rates are equal and the concentrations of reactants and products remain constant (though not necessarily equal); the equilibrium constant, K_c , for a reaction $aA + bB \rightleftharpoons cC + dD$ is $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ (using equilibrium concentrations), and its magnitude indicates the extent of reaction ($K_c \gg 1$ favours products, $K_c \ll 1$ favours reactants). K_c depends only on temperature (not on concentration, pressure, or the presence of a catalyst, since a catalyst speeds up forward and reverse rates equally); the equilibrium constant is related to the standard Gibbs free energy change by $\Delta G^\circ = -RT \ln K$, linking thermodynamic favourability (ΔG° , which also equals $\Delta H^\circ - T\Delta S^\circ$) to the position of equilibrium.

Worked example: follow the reasoning

At a certain temperature, the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ reaches equilibrium with $[H_2] = 0.020 \text{ mol dm}^{-3}$, $[I_2] = 0.020 \text{ mol dm}^{-3}$, and $[HI] = 0.16 \text{ mol dm}^{-3}$ in a fixed container. Calculate K_c at this temperature and comment on the extent of reaction.

Step 1: Write the K_c expression: $K_c = \frac{[HI]^2}{[H_2][I_2]}$.

Step 2: Substitute values: $K_c = \frac{(0.16)^2}{(0.020 \times 0.020)}$.

Step 3: $K_c = 0.0256 / 0.00040 = 64$.

Answer: $K_c = 64$ (no units, since moles of gas are equal on both sides). Since $K_c \gg 1$, the equilibrium mixture favours products (mostly HI) at this temperature.

Quick check · multiple choice

For a reaction at equilibrium, what is the effect of adding a catalyst on K_c and on the time taken to reach equilibrium?

- A K_c increases; equilibrium is reached faster
- B K_c stays the same; equilibrium is reached faster
- C K_c decreases; equilibrium is reached faster
- D K_c stays the same; equilibrium is reached at the same rate

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ has $\Delta H^\circ = +58 \text{ kJ mol}^{-1}$. At 25°C , an equilibrium mixture in a 1.00 dm^3 container contains $0.0400 \text{ mol } N_2O_4$ and $0.120 \text{ mol } NO_2$. [8]
 - (a) Write the expression for K_c for this reaction. [1]
 - (b) Calculate the value of K_c at 25°C , including units. [3]
 - (c) Predict, with a reason based on Le Chatelier's principle, whether K_c would increase, decrease or stay the same if the temperature were raised. [2]
 - (d) State the relationship between ΔG° and K_c , and use it to state whether ΔG° for this reaction is positive, negative or zero at 25°C , given that $K_c < 1$. [2]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $K_c = [\text{NO}_2]^2 / [\text{N}_2\text{O}_4]$. [1]

(b) $[\text{N}_2\text{O}_4] = 0.0400 \text{ mol dm}^{-3}$, $[\text{NO}_2] = 0.120 \text{ mol dm}^{-3}$ (1.00 dm³ container). $K_c = (0.120)^2 / 0.0400 = 0.0144/0.0400 = 0.360 \text{ mol dm}^{-3}$. [3]

(c) The forward reaction is endothermic (ΔH° positive); by Le Chatelier's principle, raising the temperature shifts the equilibrium position to favour the endothermic (forward) direction, increasing the proportion of products at equilibrium, so K_c increases. [2]

(d) $\Delta G^\circ = -RT \ln K$. Since $K_c < 1$, $\ln K_c$ is negative, so $-RT \ln K_c$ is positive; therefore ΔG° is positive (reaction is not spontaneous/favourable in the forward direction under standard conditions at 25 °C). [2]

Quick check answer

B. A catalyst increases the rate of both the forward and reverse reactions equally, so equilibrium is reached more quickly, but the equilibrium position/ K_c (which depends only on temperature) is unchanged.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

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